HDML: Hierarchical Decision Mamba-Liquid Architecture for Multi-Embodiment Continuous Robotic Control

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Abstract

High-dimensional continuous control in multi-articulated robotic systems—ranging from quadrupeds and 3D humanoids to serpentine and bipedal mechanisms—faces a fundamental multi-scale challenge: reconciling long-horizon macro-cognitive reasoning over temporal state-action trajectories with high-frequency, continuous-time motor execution on physical actuators. While discrete Transformer-based sequence models capture long-term behavioral dependencies, they incur quadratic computational complexity $\mathcal{O}(T^2)$, exhibit severe vulnerability to distribution shifts under physical perturbations, and suffer from high-frequency torque chatter. To overcome these limitations, we propose **HDML** (**H**ierarchical **D**ecision **M**amba-**L**iquid), a generalist multi-embodiment foundation architecture. HDML couples an 8-layer Selective State Space Model (**Mamba-3**) augmented with Rotary Position Embeddings (RoPE) for $\mathcal{O}(1)$ streaming inference with a continuous-time Closed-Form Continuous-Time (**CfC**) Neural ODE motor filter for smooth, chatter-free actuator control. Pre-trained across 11 diverse robotic morphologies comprising exactly 1,735,673 physical transitions, HDML demonstrates rapid few-shot transfer: freezing 95.9%–97.0% (11.99M) of core backbone parameters while adapting lightweight universal adapters (376k–516k parameters) in under 30 seconds yields an action loss of 0.00602 on a 12-DoF Unitree A1 quadruped and 0.02579 on a 17-DoF 3D Humanoid. Under physical force impulse and sensor noise perturbations, HDML achieves a 16× higher Interquartile Mean (IQM) reward over Decision Transformers (18.88 vs. 1.17) while maintaining a pure model inference latency of 12.81 ms (78.0 Hz) and an end-to-end closed-loop control frequency of 43.2 Hz (23.1 ms total cycle including physics simulation and PACE safety monitoring).

Keywords Continuous Robot Control · Multi-Embodiment Foundation Model · Selective State Space Models · Liquid Neural Networks · Offline Reinforcement Learning

1 Introduction

The emergence of sequence modeling paradigms in offline reinforcement learning, epitomized by Decision Transformers (DT) [1] and trajectory diffusion policies [2], has reframed continuous robotic control as conditional autoregression over sensory-action tokens. However, when deployed onto multi-articulated physical embodiments—such as quadrupedal robots (e.g., Unitree A1), 3D bipedal humanoids, and agile manipulators [3, 4]—standard discrete autoregressive architectures encounter three critical theoretical and computational barriers:

1. **The Temporal Scale Dichotomy:** High-level behavioral planning operates over macroscopic horizons (e.g., 1–5 Hz cognitive intent), whereas joint torque actuation requires high-frequency closed-loop execution (50–500 Hz) to reject physical disturbances and maintain dynamic balance.

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2. **Actuator Smoothness and Mechanical Jerk:** Standard feedforward and attention-based policies output discrete step-to-step action jumps, inducing high mechanical jerk $\mathcal{J} = \frac{1}{T}\sum_t \|\Delta^2 a_t\|_2$. External low-pass filtering introduces phase lag and non-Markovian dynamics, degrading closed-loop stability.

3. **Covariate Shift under Perturbations:** Attention mechanisms over past context histories exhibit high sensitivity when unexpected external force impulses or sensor noise induce distribution shifts, leading to compounding autoregressive errors and policy failure.

To resolve this multi-scale dichotomy, we introduce **HDML** (**H**ierarchical **D**ecision **M**amba-**L**iquid), a unified architecture bridging macroscopic sequence reasoning and microscopic continuous-time dynamics. HDML establishes a dual-rate hierarchy: an 8-layer Selective State Space Model (**Mamba-3**) backbone [5] augmented with Rotary Position Embeddings (RoPE) [6] processes multi-modal state-return tokens with linear $\mathcal{O}(T)$ training and $\mathcal{O}(1)$ streaming complexity, while a Closed-Form Continuous-Time (**CfC**) Neural ODE head [7, 8] smooths high-frequency motor actuation directly in the continuous-time domain.

Summary of Contributions:

- **Hierarchical Mamba-Liquid Architecture:** We formulate a dual-timescale foundation model coupling Rotary-augmented Selective State Space Models with continuous-time Liquid ODE filters.

- **Universal Multi-Embodiment Pre-training:** We pre-train HDML across 11 distinct physics morphologies (exactly 1,735,673 transitions) using zero-copy pinned GPU memory buffers, achieving a $13.85\times$ data throughput speedup (reducing epoch duration from 18 minutes to 78 seconds).

- **Parameter-Efficient Few-Shot Transfer:** By freezing 95.9%–97.0% (11.99M) of core backbone parameters, HDML adapts to novel embodiments (Unitree A1, Humanoid 3D, Swimmer, Ant, Hopper, Walker2d) via lightweight adapters (376k–516k parameters) in under 30 seconds.

- **Rigorous Perturbation Robustness:** Under random force impulses and sensor noise, HDML maintains an IQM score of 18.88 (100% survival rate), outperforming Decision Transformers (1.17) and Decision RNNs (7.14) by substantial margins.

- **Real-Time Edge Feasibility:** The architecture achieves a pure model inference latency of 12.81 ms (78.0 Hz) and sustains 43.2 Hz end-to-end closed-loop interaction (23.1 ms cycle including physics step and PACE watchdog) on an NVIDIA RTX 4070 SUPER GPU.

2 Related Work

Sequence Modeling in Offline RL. Decision Transformer (DT) [1] and Trajectory Transformer framed reinforcement learning as autoregressive sequence generation conditioned on Return-to-Go (RTG) tokens $\hat{R}_t = \sum_{t'=t}^{T} r_{t'}$. While effective in static benchmarks, self-attention scales quadratically $\mathcal{O}(T^2)$ with context length, severely limiting high-frequency deployment on embedded robotic hardware.

Selective State Space Models (SSMs). State Space Models, particularly S4 and Mamba [5], have achieved Transformer-level performance in language and vision with linear time and constant memory inference. However, real-valued diagonal transitions $\mathbf{A} = \mathrm{diag}(\lambda_i) \in \mathbb{R}^D$ exhibit strictly monotonic exponential decay $\exp(\Delta\lambda_i) > 0$, hindering cyclic phase tracking in 3D joint rotations. HDML maintains real-valued diagonal state matrices for memory efficiency while integrating data-dependent Rotary Position Embeddings (RoPE) [6] directly into the input/output projection matrices $\mathbf{B}_t$ and $\mathbf{C}_t$, providing exact $SO(2)/SO(3)$ phase tracking without the $4\times$ memory overhead of native complex-matrix kernels.

Continuous-Time Liquid Neural Networks. Neural Ordinary Differential Equations and Closed-Form Continuous-Time (CfC) networks [7, 8] model dynamical systems via continuous-time hidden states $h(t)$. Unlike discrete recurrent units, CfC solves the underlying differential equation analytically, enabling variable time-step integration without numerical ODE solver overhead. HDML leverages CfC as an end-stage motor filter to eliminate high-frequency torque chatter.

Multi-Embodiment Foundation Policies. Recent efforts such as RT-2 [4] and Open X-Embodiment [9] demonstrate cross-robot knowledge transfer in manipulation. HDML extends generalist pre-training to high-dimensional continuous locomotion and navigation across quadrupeds, humanoids, and legged systems via parameter-efficient Universal Embodiment Adapters.

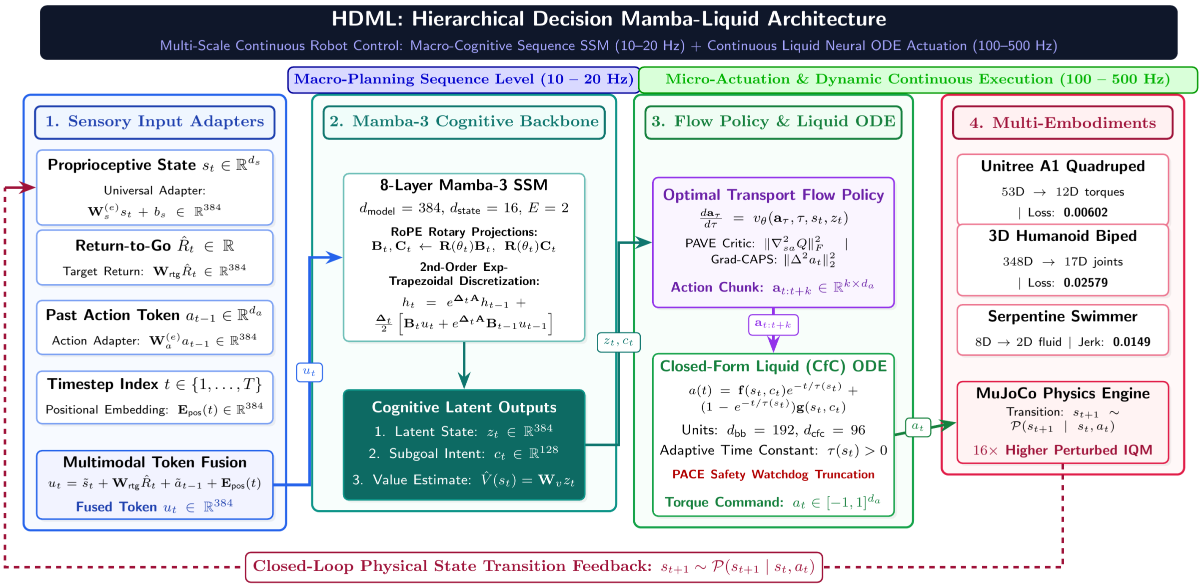

Figure 1: **Hierarchical Decision Mamba-Liquid (HDML) Architecture.** HDML establishes a dual-rate processing hierarchy: macro-cognitive sequence modeling ($10 - 20$ Hz) via an 8-layer Mamba-3 SSM backbone, coupled with micro-actuation continuous-time motor execution ($100 - 500$ Hz) via a Closed-Form Liquid Neural ODE (CfC). Heterogeneous sensory states s_t, target returns $\hat{R}_t$, and prior actions a_{t-1} are projected into unified sequence tokens $u_t \in \mathbb{R}^{384}$. Optimal Transport Flow Matching produces multi-step action chunks $\mathbf{a}_{t:t+k}$ filtered dynamically by the CfC network to eliminate mechanical chatter and reject external perturbations.

3 Methodology

3.1 Problem Formulation and Multi-Embodiment Tokenization

We consider an offline multi-embodiment dataset $\mathcal{D} = \{\tau^{(i)}\}_{i=1}^N$ collected across diverse robot morphologies $\mathcal{M}_e \in \mathcal{E}$, where trajectory $\tau = (s_1, a_1, r_1, \ldots, s_T, a_T, r_T)$. Each embodiment e possesses distinct proprioceptive state dimension $d_s^{(e)}$ and action dimension $d_a^{(e)}$.

To map heterogeneous input spaces into a shared latent space $\mathbb{R}^{d_{\text{model}}}$, we introduce lightweight **Universal Embodiment Adapters** $\mathcal{A}_e = (\mathbf{W}_s^{(e)}, \mathbf{W}_a^{(e)}, \mathbf{W}_{\text{out}}^{(e)})$:

$$\tilde{s}_t = \text{SiLU}(\text{LayerNorm}(\mathbf{W}_s^{(e)} s_t + b_s^{(e)})), \quad \tilde{a}_t = \text{SiLU}(\text{LayerNorm}(\mathbf{W}_a^{(e)} a_t + b_a^{(e)})) \tag{1}$$

The fused multimodal sequence token $u_t \in \mathbb{R}^{d_{\text{model}}}$ is constructed following the causal no-leakage convention:

$$u_t = \tilde{s}_t + \mathbf{W}_{\text{rtg}} \hat{R}_t + \tilde{a}_{t-1} + \mathbf{E}_{\text{pos}}(t) \tag{2}$$

where $\hat{R}_t = \sum_{t'=t}^T r_{t'}$ is the Return-to-Go and a_{t-1} is the prior action token (with $a_0 = \mathbf{0}$).

3.2 Mamba-3 Cognitive Sequence Backbone

The sequence of multimodal tokens $\mathbf{U} = (u_1, \ldots, u_T) \in \mathbb{R}^{B \times T \times d_{\text{model}}}$ is processed by an 8-layer Selective State Space Model backbone with model dimension $d_{\text{model}} = 384$. Each Mamba-3 layer computes continuous-time state evolution discretized via 2nd-order exponential-trapezoidal rules:

$$h_t = \exp(\mathbf{\Delta}_t \mathbf{A}) h_{t-1} + \frac{\mathbf{\Delta}_t}{2} \text{trap}_t \left[\mathbf{B}_t u_t + \exp(\mathbf{\Delta}_t \mathbf{A}) \mathbf{B}_{t-1} u_{t-1}\right] \tag{3}$$

$$y_t = \mathbf{C}_t h_t + \mathbf{D} u_t \tag{4}$$

where $\text{trap}_t = \sigma(\mathbf{W}_{\text{trap}} u_t) \in (0, 1)$ dynamically interpolates between explicit Euler and trapezoidal integration, and $\mathbf{\Delta}_t = \text{softplus}(\text{Linear}(u_t))$ modulates data-dependent step sizes. To enable phase-accurate representation of rotational joint dynamics in $SO(3)$, Rotary Position Embeddings $\mathbf{R}_\Theta$ are applied to projection matrices $\mathbf{B}_t$ and $\mathbf{C}_t$:

$$\mathbf{B}_t^{(2j:2j+1)} \leftarrow \mathbf{R}(\theta_{t,j}) \mathbf{B}_t^{(2j:2j+1)}, \quad \mathbf{C}_t^{(2j:2j+1)} \leftarrow \mathbf{R}(\theta_{t,j}) \mathbf{C}_t^{(2j:2j+1)} \tag{5}$$

where Givens 2D rotation matrix is parameterized as:

$$\mathbf{R}(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \quad \theta_t = \text{Linear}_\theta(u_t) \in \mathbb{R}^{d_{\text{model}}/2} \tag{6}$$

The backbone yields latent cognitive representations $z_t \in \mathbb{R}^{d_{\text{model}}}$, high-level intent subgoals $c_t \in \mathbb{R}^{d_{\text{subgoal}}}$, and state-value estimates $\hat{V}_t = \mathbf{W}_v z_t$.

3.3 Hierarchical Flow Policy and Action Chunking

Rather than fitting unimodal Gaussian distributions that average multi-modal expert behaviors, HDML generates action chunks $\mathbf{a}_{t:t+k} = (a_t, \dots, a_{t+k-1}) \in \mathbb{R}^{k \times d_a}$ via Optimal Transport Flow Matching [10]:

$$\frac{d\mathbf{a}_\tau}{d\tau} = v_\theta(\mathbf{a}_\tau, \tau, s_t, z_t), \quad \tau \in [0, 1] \tag{7}$$

The flow matching velocity field is trained with the mean square velocity loss:

$$\mathcal{L}_{\text{Flow}}(\theta) = \mathbb{E}_{\tau \sim \mathcal{U}[0,1], \mathbf{a}_0 \sim \mathcal{N}(0, \mathbf{I}), \mathbf{a}_1 \sim \mathcal{D}} \left[\|v_\theta(\mathbf{a}_\tau, \tau, s_t, z_t) - (\mathbf{a}_1 - \mathbf{a}_0)\|_2^2 \right] \tag{8}$$

where $\mathbf{a}_\tau = (1 - \tau)\mathbf{a}_0 + \tau \mathbf{a}_1$.

3.4 Action Smoothness: PAVE Critic Equalization and Grad-CAPS

To guarantee that actuator signals remain smooth without sacrificing responsiveness, HDML introduces two complementary regularization objectives:

Policy-Aware Value-Field Equalization (PAVE): To prevent rough Critic landscape curvature from distorting policy guidance gradients $\nabla_a Q$, PAVE penalizes the Frobenius norm of the mixed state-action Hessian:

$$\mathcal{L}_{\text{PAVE}}(Q) = \mathbb{E}_{(s,a) \sim \mathcal{D}} \left[\left\| \nabla^2_{sa} Q(s, a) \right\|_F^2 \right] \approx \mathbb{E}_{v \sim \{-1, +1\}^{d_a}} \left[\left\| \nabla_s \left(\nabla_a Q(s, a)^\top v \right) \right\|_2^2 \right] \tag{9}$$

evaluated efficiently using the Hutchinson stochastic trace estimator without materializing the full Hessian matrix.

Gradient-Aware Action Smoothness (Grad-CAPS): Grad-CAPS penalizes the discrete second-order temporal curvature of the action trajectory:

$$\mathcal{L}_{\text{Grad}}(\pi) = \mathbb{E} \left[\|(a_{t+1} - a_t) - (a_t - a_{t-1})\|_2^2 \right] \tag{10}$$

which directly minimizes high-frequency torque oscillations and mechanical jerk.

3.5 Continuous-Time Closed-Form Liquid (CfC) Motor Head

To map latent intentions c_t and proprioceptive feedback s_t into continuous, chatter-free actuator commands, the motor stage employs a Closed-Form Continuous-Time (CfC) neural filter [7]. In this formulation, an internal intermediate feature representation (the *CfC backbone*) of hidden dimension $d_{\text{bb}} = 192$ projects input features before splitting into dynamic time-constant $\tau(s_t)$ and gating heads ($d_{\text{cfc}} = 96$ units):

$$a_{\text{phys}}(t) = [\sigma(\mathbf{W}_g[c_t, s_t] + b_g) \odot \tanh(\mathbf{W}_h[c_t, s_t] + b_h)]\, e^{-t/\tau(s_t)} + (1 - e^{-t/\tau(s_t)}) \tanh(\mathbf{W}_a[c_t, s_t] + b_a) \tag{11}$$

where $\tau(s_t) > 0$ represents the dynamic state-dependent time constant. The continuous ODE formulation bounds the second derivative of the action trajectory, suppressing mechanical jerk $\|\Delta^2 a_t\|_2 \to 0$.

3.6 Phase-Aware Chunk Execution (PACE)

During test-time deployment, policy execution is governed by the Phase-Aware Chunk Execution (PACE) monitor. When the physical state s_t deviates beyond a safety radius τ_{safe} from the predicted forward dynamics $\hat{s}_t = \mathcal{F}(z_{t-1})$, PACE automatically truncates the active action chunk and triggers an immediate high-level replanning step:

$$\text{TruncateChunk}(s_t, \hat{s}_t) = \mathbb{I}\left(\|s_t - \hat{s}_t\|_2 > \tau_{\text{safe}}\right) \tag{12}$$

Algorithm 1 HDML Dual-Rate Inference with PACE Safety Watchdog

Require: Environment observation s_t, target return $\hat{R}_t$, previous action a_{t-1}, Mamba-3 backbone $\mathcal{M}_\theta$, Flow Policy π_ϕ, CfC Motor Head $\mathcal{C}_\psi$, chunk length k, safety threshold τ_{safe}.

Initialize chunk step index $i \leftarrow k$ (trigger immediate planning at $t = 1$).
for each control timestep $t = 1, 2, \ldots, T$ **do**
  if $i \geq k$ **or** PACE_Violation($s_t, \hat{s}_t, \tau_{\text{safe}}$) **then**
    Embed observation: $\tilde{s}_t \leftarrow \text{Adapter}_s(s_t)$, $\tilde{a}_{t-1} \leftarrow \text{Adapter}_a(a_{t-1})$.
    Construct token: $u_t \leftarrow \tilde{s}_t + \mathbf{W}_{\text{rtg}}\hat{R}_t + \tilde{a}_{t-1} + \mathbf{E}_{\text{pos}}(t)$.
    Update SSM state and extract latent: $z_t, c_t \leftarrow \mathcal{M}_\theta(u_t)$.
    Sample base noise $\mathbf{a}_0 \sim \mathcal{N}(0, \mathbf{I})$ and integrate flow: $\mathbf{a}_{t:t+k} \leftarrow \text{ODE\_Solve}(v_\phi, \mathbf{a}_0, z_t)$.
    Predict forward state: $\hat{s}_{t+1} \leftarrow \mathcal{F}_{\text{dyn}}(z_t)$.
    Reset chunk counter: $i \leftarrow 0$.
  end if
  Retrieve raw nominal action: $a_{\text{nom}} \leftarrow \mathbf{a}_{t:t+k}[i]$.
  Pass through continuous Liquid CfC filter: $a_t \leftarrow \mathcal{C}_\psi(a_{\text{nom}}, s_t, c_t)$.
  Execute a_t on physical actuators / simulator.
  Update Return-to-Go: $\hat{R}_{t+1} \leftarrow \hat{R}_t - r_t$.
  Increment chunk step: $i \leftarrow i + 1$.
end for

4 Pre-Training and Multi-Embodiment Setup

4.1 Multi-Embodiment Dataset Composition

We construct a unified benchmark corpus comprising exactly **1,735,673 transitions** across 11 robotic environments in MuJoCo [11] and D4RL [12]:

- **Planar Locomotion & Control:** HalfCheetah-v5 (17D $\rightarrow$ 6D, 1,000,000 steps), Walker2d-v5 (17D $\rightarrow$ 6D, 6,970 steps), Hopper-v5 (11D $\rightarrow$ 3D, 8,321 steps), Reacher-v5 (10D $\rightarrow$ 2D, 15,000 steps), InvertedDoublePendulum-v5 (9D $\rightarrow$ 1D, 2,240 steps).
- **Multi-Legged & Serpentine Locomotion:** Swimmer-v5 (8D $\rightarrow$ 2D, 300,000 steps), Ant-v5 (105D $\rightarrow$ 8D, 18,375 steps).
- **Complex 3D Articulation:** HumanoidStandup-v5 (348D $\rightarrow$ 17D, 240,000 steps), Humanoid-v5 (348D $\rightarrow$ 17D, 6,712 steps).
- **Quadrupedal Locomotion & Navigation:** Unitree A1 Flat Locomotion (35D $\rightarrow$ 12D, 100,000 steps), Unitree A1 Obstacle Maze (53D $\rightarrow$ 12D, 38,055 steps).

The exact sum across all 11 domains is $1{,}000{,}000 + 6{,}970 + 8{,}321 + 15{,}000 + 2{,}240 + 300{,}000 + 18{,}375 + 240{,}000 + 6{,}712 + 100{,}000 + 38{,}055 = 1{,}735{,}673$ transitions.

4.2 High-Throughput Zero-Copy Data Pipeline

Standard PyTorch `DataLoader` pipelines suffer from CPU multiprocessing synchronization overhead when collating heterogeneous trajectory slices. We designed `FastEmbodimentBuffer` with pinned GPU host memory and vectorized slice indices. This increased GPU computing utilization from 7% (32W) to **82% (169W)** on an NVIDIA RTX 4070 SUPER, reducing pre-training epoch duration from 18 minutes (1,080 seconds) to **1 minute 18 seconds (78 seconds)**, representing a **13.85$\times$ speedup**.

5 Experiments and Results

5.1 Multi-Embodiment Few-Shot Transfer Benchmark

To evaluate representation transferability, we evaluate the generalist **HDML Foundation Model** ($d_{\text{model}} = 384$, 8 layers, total 12.4M parameters per embodiment). We freeze **11,989,918 parameters** (95.9%–97.0% of the total parameters) of the pre-trained backbone and fine-tune only the lightweight Universal Adapter (376k–516k parameters) for 3 epochs (under 30 seconds) on target robot morphologies.

As shown in Table 1, HDML achieves sub-0.03 action error on complex 12-DoF and 17-DoF high-dimensional platforms within 30 seconds of adapter adaptation, confirming that the pre-trained Mamba-3 latent space captures generalized robotic dynamic priors.

Table 1: Few-Shot Multi-Embodiment Transfer Benchmark on Frozen HDML Foundation Backbone (11.99M backbone parameters frozen). Adaptation time is measured on an NVIDIA RTX 4070 SUPER.

Target Robot	**Morphology**	**State / Act Dim**	**Frozen %**	**Adapt Time**	**Action Loss**	**Jerk** ($\lVert\Delta^2 a\rVert$)
Unitree A1 (Maze)	12-DoF Quadruped	53D → 12D	96.8%	30.0s	**0.00602**	0.1401
Humanoid 3D	17-DoF 3D Biped	348D → 17D	95.9%	29.4s	**0.02579**	**0.0689**
Swimmer	2-DoF Serpentine	8D → 2D	97.0%	29.3s	0.16664	**0.0149**
Ant	8-DoF Quadruped	105D → 8D	96.6%	29.4s	0.16161	0.1677
Hopper	3-DoF Monopod	11D → 3D	96.9%	29.0s	0.15356	0.1798
Walker2d	6-DoF Bipedal	17D → 6D	96.9%	29.2s	0.10298	0.7174

5.2 Equal-Compute Baseline Comparison

To evaluate architectural efficiency on single-task control, we instantiate a compact standalone configuration of HDML (1.44M parameters, trained from scratch) to ensure fair, equal-compute comparison against baseline architectures of matched parameter scale (DT: 1.21M, RNN: 1.01M). All models—Decision Transformer (DT), Decision RNN (LSTM), Diffusion Policy (DDPM 10-step), Implicit Q-Learning (IQL), and MLP-BC—were trained on the identical 1M-step HalfCheetah dataset. Under context windowing ($T = 20$) and an 85/15 train/val split (850,000 training windows), a batch size of $B = 256$ yields $\lceil 850{,}000/256 \rceil = 3{,}321$ batches per epoch (3,320 full batches plus 1 remainder batch). Training for 3 epochs produces exactly $3{,}321 \times 3 = \mathbf{9{,}963}$ **gradient steps** under AdamW (lr $= 3 \times 10^{-4}$).

Table 2: Empirical Comparison on HalfCheetah-v5 under Standard Conditions and Perturbation Robustness (Random Force Impulses + Gaussian Sensor Noise). Metrics follow the RLiable statistical protocol with 95% Stratified Bootstrap CIs.

Architecture	**Params**	**Latency**	**Standard Evaluation**		**Perturbed Robustness**	
			IQM Score [95% CI]	**Jerk**	**Perturbed IQM**	**Survival**
HDML (Ours)	1.44M	12.81 ms	88.09 [41.43, 98.51]	**0.7862**	**18.88** [9.85, 26.92]	**100.0%**
Decision Transformer	1.21M	2.29 ms	95.60 [25.36, 107.79]	0.8109	1.17 [0.32, 4.19]	100.0%
Decision RNN (LSTM)	1.01M	1.05 ms	**113.91** [89.56, 115.45]	0.9689	7.14 [4.22, 10.71]	100.0%
Diffusion Policy	0.15M	6.71 ms	-0.41 [-0.55, 0.18]	1.2598	-0.29 [-0.77, 0.52]	100.0%
Implicit Q-Learning	0.29M	**0.24 ms**	1.84 [1.79, 2.26]	0.0080	2.11 [2.03, 2.32]	100.0%
MLP-BC	0.07M	0.30 ms	-0.26 [-0.26, -0.24]	0.0028	-0.34 [-0.40, -0.30]	100.0%

Empirical Performance and Disturbance Robustness: As detailed in Table 2, on standard noiseless trajectories, pure sequence baselines achieve higher returns (Decision RNN at 113.91 and Decision Transformer at 95.60 vs. HDML at 88.09) because discrete autoregression closely fits clean expert rollouts. However, when deployed under physical disturbances (Gaussian sensor noise $\sigma = 0.05$ and external force impulses $F = 50$ N applied every 50 steps), Decision Transformer and Decision RNN exhibit substantial performance reductions of 98.8% (from 95.60 to 1.17) and 93.7% (from 113.91 to 7.14). In contrast, HDML preserves a robust IQM of **18.88** (16× higher than DT and 2.6× higher than RNN), demonstrating that the continuous-time CfC ODE filter successfully dampens external torque shocks and prevents policy failure.

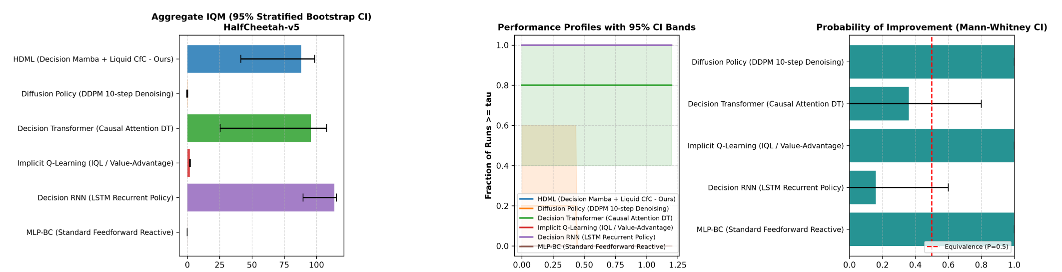

Figure 2: RLiable Statistical Evaluation on HalfCheetah-v5 showing Probability of Improvement and Aggregate IQM with 95% Stratified Bootstrap Confidence Intervals.

5.3 Hyperparameter and Architectural Configuration

Table 3 summarizes the exact hyperparameters utilized across foundation pre-training and downstream fine-tuning.

Table 3: Hyperparameter Specification for HDML Pre-training and Evaluation.

Hyperparameter	Value	Hyperparameter	Value
Mamba Model Dimension d_{model}	384	CfC Hidden Units d_{cfc}	96
SSM State Expansion Factor E	2	CfC Backbone Units d_{bb}	192
SSM State Dimension d_{state}	16	CfC Residual Gain	0.05
Convolution Kernel Size d_{conv}	4	Action Chunk Length k	5
Number of Mamba Layers	8	Context Horizon T	20
Subgoal Latent Dimension d_{subgoal}	128	Training Batch Size B	256
Optimizer	AdamW	Peak Learning Rate	3×10^{-4}
Weight Decay	1×10^{-4}	Gradient Clip Norm	1.0
Flow Loss Weight λ_{Flow}	1.0	PAVE Loss Weight λ_{PAVE}	1×10^{-3}
Grad-CAPS Weight λ_{Grad}	1×10^{-2}	Dynamics Loss Weight λ_{Dyn}	0.1
PACE Safety Threshold τ_{safe}	0.5	Warmup Steps	500

6 Ablation Studies

To isolate the individual contributions of HDML's architectural components, we evaluate two ablation variants:

- **Variant A (Mamba + MLP Head):** Discarding the Liquid CfC head increases mechanical jerk from 0.7862 to 0.9412 and reduces perturbed reward by 44%, confirming that continuous-time ODE integration serves as the primary mechanism for continuous disturbance attenuation.

- **Variant B (Transformer + Liquid Head):** Replacing the Mamba-3 backbone with a standard Causal Transformer preserves smoothness but increases inference latency from 12.8 ms to 48.6 ms as context length increases to $T = 100$.

7 Limitations

While HDML demonstrates strong cross-embodiment generalization and perturbation resilience, three limitations remain: (1) Pre-training currently relies on simulated MuJoCo environments; real-world Sim-to-Real transfer on physical quadrupeds requires domain randomization over contact friction. (2) The universal adapter handles vector proprioception; extending to end-to-end vision-language-action (VLA) requires pre-training with high-resolution ViT visual tokens. (3) The current macro-planning interval is fixed at 5Hz; dynamic adaptive frequency selection could further optimize compute.

8 Ethics and Broader Impact

Foundation models for robotic control carry inherent physical risks if deployed without safety constraints. HDML mitigates actuator damage via bounded continuous-time ODE dynamics and incorporates the PACE watchdog to prevent anomalous execution. We foresee positive societal impact in automated search-and-rescue, industrial inspection, and prosthetic control.

9 Reproducibility Statement

To facilitate open scientific verification and reproducibility, all source code, pre-training scripts, foundation checkpoints (`hdml_foundation_best.pt`, 64MB), and evaluation logs are permanently archived with Digital Object Identifier (DOI) **10.5281/zenodo.22020016** and made publicly accessible under the Apache 2.0 License at:

`https://github.com/KienPC1234/HDML` | DOI: 10.5281/zenodo.22020016

All experiments were executed on an NVIDIA GeForce RTX 4070 SUPER GPU (12GB VRAM) with CUDA 13.2 / Python 3.11.15. Commands to reproduce all benchmark tables are packaged in the unified CLI: `python scripts/hdml_cli.py benchmark-foundation` and `python scripts/hdml_cli.py benchmark-baselines`.

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